INCH-POUND

MIL-DTL-63090B(AR)
15 November 2022
SUPERSEDING
MIL-F-63090A(AR)
21 APRIL 1981

MILITARY SPECIFICATION

FIRING DEVICE, MULTI-PURPOSE, M142
PARTS, ASSEMBLY, AND PACKING

Reactivated after 9 November 2022 and may be used for be used for new and existing acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of firing device, designated as Firing Device, Multi-Purpose, M142.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those specified in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to: usarmy.picatinny.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

AMSC N/A FSC 1375

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-20444 - Primer, Percussion, M42, Parts For and Loading Assembly and Packing

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

U.S. ARMY DEVCOM ARMAMENTS CENTER DRAWINGS

9296865	-	Firing device demolition multi-purpose: M142
9296866	-	Coupling body assembly
9296867	-	Cable reel assembly
9296868	-	Switch assembly
9296869	-	Pin and cord assembly
9296871	-	Mounting hardware
9296872	-	Nut, coupling
9296874	-	Instruction sheet
9296875	-	Plate, sear
9296876	-	Body, coupling
9296877	-	Body
9296878	-	Reel, cable
9296880	-	Packing, preformed
9296881	-	Spring, firing, device, demolition
9296882	-	Pin, square head loop
9296883	-	Pin, round head loop
9296884	-	Tension release attachment
9296885	-	Striker, firing device demolition
9311138	-	Packing and marking for box, wood, wirebound for firing device demolition, multipurpose, M142 in box, ammunition, M19A1
9311139	-	Liner, for box, ammunition, M19A1
9311140	-	Box, wood, wirebound
9311141	-	Packing and marking for box, ammunition, M19A1, with firing device, demolition, multipurpose, M142
9335567	-	Grommet
9335590	-	Container, lid
9335591	-	Container, bottom

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

9349653	-	Test Cap Blasting Non-Electric for M142 F.D.
9352613	-	Plastic container and lid
12589971	-	Clip, retaining

(Copies of these drawings may be requested by email at usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Materials. Materials shall be in accordance with the drawings and specifications cited in 4.3.2.1-4.3.3.5.

3.3 Components and assemblies. The components and assemblies shall comply with all requirements specified on 9296865.

3.4 Functioning. The firing device shall not prematurely release the firing pin and shall mechanically initiate the primer providing actuation by pull, tension release, pressure, and pressure release as follows.

3.4.1 Pull. The firing device shall function the primer when a load of not less than 4.0 pounds (lbs) to not greater than 9.0 lbs is applied to the trip wire in a direction perpendicular to the longitudinal axis of the firing device body.

3.4.2 Tension release. The firing device shall function the primer when a tension of 3 lbs ± 0.25 lbs in the trip wire applied to the tension release attachment is released upon cutting the trip wire or flipping it off a pin.

3.4.2.1 Position of tension release attachment. The tension of 3 lbs ± 0.25 lbs shall place the tension release attachment to the set position where the gooseneck arm (Leg A per Drawing 9296884) is adjacent to the extreme tip of the straight arm (Leg B per Drawing 9296884).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

3.4.3 Pressure. The firing device shall function the primer when a varying load of not less than 15 lbs to not greater than 25 lbs is applied to the sear platform.

3.4.4 Pressure release. The firing device shall function at not greater than 2 lbs when a load of not less than 4 lbs is applied to the sear platform and then gradually reduced.

3.5 Water immersion. The firing device shall be capable of operating and meeting the requirements of 3.4.1 and 3.4.3 when submerged in not less than 6 inches of water.

3.6 Cold temperature. The firing device shall be capable of operating and meeting the requirements of 3.4.1, 3.4.3, and 3.4.4 when conditioned at not greater than -45°F for 16 hours.

3.7 Check test for possible deterioration of primers. If the total time between original acceptance of any primer lot and the assembly of that lot into the coupling body of firing device exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lots shall be subjected to and must satisfactorily pass the check test for deterioration.

3.8 Body, coupling web. The web in the body coupling shall not rupture, nor show evidence of leakage when subjected to an air pressure of 5-10 pounds per square inch (psig) for not less than 15 seconds.

3.9 Workmanship. All parts and assemblies shall be fabricated and finished and assembly operations shall be correctly performed. They shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity, warpage, burn marks, checks, chipped edges, blisters, excess flash, dirt, grease, rust, salt deposits, visible raised cement seams, solder splash, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Method of Verification	Classes of Verification
1- Analysis 2- Demonstration (end item test) 3- Examination 4- Test	A - First Article B - Conformance

Section 3 Requirements	Description	Verification Method				Verification Class		Section 4
		1	2	3	4	A	B	
3.1.1	First Article	X	X	X	X	X		4.2
3.1.2	Conformance	X	X	X	X		X	4.3
3.2	Materials	X		X	X	X	X	4.3
3.3	Components and assemblies			X	X	X	X	4.3
3.4	Functioning		X			X	X	4.4
3.5	Water immersion		X			X	X	4.4.2
3.6	Cold temperature		X			X	X	4.4.3
3.7	Check test for primers			X	X	X	X	4.4.4
3.8	Body, coupling web		X	X	X	X	X	4.4.5
3.9	Workmanship	X		X	X	X	X	4.3
5.1	Packaging	X		X	X	X	X	4.3.2.19-24

4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Submission. When specified, a first article sample shall be submitted for evaluation in accordance with the provisions of 4.2.2 and Table II.

4.2.2 Inspections to be performed. The first article samples shall be subjected to all of the examinations and tests specified in the specification and inspected for compliance with Table II and all of the requirements of this specification and applicable drawings, including all unlisted characteristics on those drawings.

4.2.3 Rejection. The first article sample shall be rejected if any of the inspections/tests fail to meet specification or drawing requirements. The Government reserves the right to terminate inspection or test upon failure of any sample to comply with any requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

TABLE II. First article inspection.

Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Parts and sub-assemblies			
Examination of defects (Dwg. 9296865)	25 (1/)	3.2	4.3.2.1 thru 4.3.2.18, and 4.3.2.24
Examination of defects (Dwg. 9311138)	2 (2/)	3.2, 3.3	4.3.2.19 thru 4.3.2.23
Plastic parts			
Examination of defects (Dwgs. 9296872, 9296876, 9296875, 9296877, 9352613)	5(3/)	3.2	4.3.3.1 thru 4.3.3.5
Firing device assemblies (Dwg. 9296865)	150 total	3.3	4.4
Functioning			4.4.1
Group A			
Pull	25	3.3.1	4.4.1.1
Tension release	25	3.3.2	4.4.1.2
Pressure	25	3.3.3	4.4.1.3
Pressure release	25	3.3.4	4.4.1.3
Group B (water immersion)			
Pull	10	3.3.1 & 3.4	4.4.2
Pressure	10	3.3.3 & 3.4	4.4.2
Group C (cold temp.)			
Pull	10	3.3.1 & 3.5	4.4.3
Pressure	10	3.3.3 & 3.5	4.4.3
Pressure release	10	3.3.4 & 3.5	4.4.3
Check test for primers (if applicable)	900	3.7	4.4.4
Body, coupling web	25	3.8	4.4.5
(1/) 25 each and every component and sub-assembly (2/) 2 each and every component and sub-assembly (3/) 5 each molded parts from each cavity			

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-STD-1916. In addition, inspection lots of firing devices shall contain:

a. Parts from not more than one lot interfix number from one manufacturer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

b. Primers from not more than one lot from one manufacturer.

4.3.2 Examination. Unless otherwise specified in the Conformance Inspection, sampling plans for major and minor defects shall be in accordance with MIL-STD-1916, Verification Level IV for Major defects and Verification Level II for Minor defects. Function test samples shall be in accordance with Tables III and IV. Unless otherwise noted, any assembly, subassembly, material, or component which fails to comply with the conformance inspection requirements shall reject the batch/lot.

4.3.2.1 Grommet.				Drawing Number 9335567 Next Higher Assembly 9296866
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Outside diameter	IV	3.2	Gage
102	Inside diameter	IV	3.2	Gage
103	Molding defects (see 6.5)	IV	3.2	Visual
Minor				
201	Length of flat head	II	3.2	Gage
202	Diameter of groove	II	3.2	Gage
203	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.2 Packing, preformed.				Drawing Number 9296880 Next Higher Assembly 9296866
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Inside diameter	IV	3.2	Gage
102	Molding defects (see 6.5)	IV	3.2	Gage
Minor				
201	Outside diameter	II	3.2	Gage
202	Thickness	II	3.2	Gage
203	Evidence of poor workmanship	II	3.9	Visual

4.3.2.3 Coupling body assembly.				Drawing Number 9296866 Next Higher Assembly 9296865
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Body coupling web pressure test prior to installing primer	IV	3.8	4.4.5
102	Primer missing or incorrectly installed	IV	3.2	Visual
103	Body coupling face or primer not lacquered	IV	3.2	Visual
104	Grommet missing	IV	3.2	Visual
105	Coupling nut missing	IV	3.2	Visual
106	Any packing component missing	IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.4 Reel, cable.				Drawing Number 9296878 Next Higher Assembly 9296865
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major	None defined			
Minor				
201	Outside diameter	II	3.2	Gage
202	Incomplete form 1/	II	3.2	Visual
203	Length of reel	II	3.2	Gage
204	Evidence of poor workmanship	II	3.9	Visual
(1/) Incomplete form shall consist of bubbles, voids, and insufficient material.				

4.3.2.5 Cable reel assembly.				Drawing Number 9296867-1 or 9296867-2 Next Higher Assembly 9296865
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Wire missing	IV	3.2	Visual
Minor				
201	Coiled wire not secured with lacquer (if applicable)	II	3.2	Visual
202	Loose end of wire not properly secured	II	3.2	Visual
203	Wire length (destructive test)	II	3.2	Gage
204	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.6 Tension release attachment.				Drawing Number 9296884 Next Higher Assembly 9296865
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Internal loop width on leg B	IV	3.2	Gage
102	Height of straight arm B	IV	3.2	Gage
103	Length of section opposite straight arm	IV	3.2	Gage
104	Height of step opposite straight arm	IV	3.2	Gage
105	Load test of tension release attachment	IV	3.2	Gage
Minor				
201	Oxide finish missing	II	3.2	Visual
202	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.7 Switch assembly.				Drawing Number 9296868 Next Higher Assembly 9296865
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical				
1	Safety pin not in place or legs not bent	100% + VL-VII	3.2	AAIE (1/)
2	Firing pin released against safety pin with all components (pins) present	100% + VL-VII	3.2	AAIE (1/)
Major				
101	Cord missing from assembly	IV	3.2	Visual
102	Square loop or round loop pins missing	IV	3.2	Visual
103	Alternative safety pin holes not grease packed	IV	3.2	Visual
104	Grease missing or incomplete	IV	3.2	Visual
105	Striker assembly missing	IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	II	3.9	Visual
(1/) Automated Acceptance Inspection Equipment (see 6.5)				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.8 Mounting hardware.				Drawing Number 9296871 Next Higher Assembly 9296865
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Any screw or nail missing	IV	3.2	Visual
Minor				
201	Nails or screws not properly secured	II	3.2	Visual
202	Tape ends not folded back	II	3.2	Visual
203	Evidence of poor workmanship	II	3.9	Visual

4.3.2.9 Striker, firing device, demolition.				Drawing Number 9296885 Next Higher Assembly 9296868
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Length of point	IV	3.2	Gage
102	Profile of point	IV	3.2	Gage
103	True position of four tabs with outside diameter	IV	3.2	Gage
104	Outside diameter of striker	IV	3.2	Gage
105	Diameter of striker undercut	IV	3.2	Gage
106	Length of undercut	IV	3.2	Gage
107	Length from undercut to tabs	IV	3.2	Gage
Minor				
201	Overall length incorrect	II	3.2	Gage
202	Angle at undercut	II	3.2	Gage
203	Thickness of tabs, 4 places	II	3.2	Gage
204	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.10 Pin, round loop head.				Drawing Number 9296883 Next Higher Assembly 9296868
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major	None defined			
Minor				
201	Length to loop	II	3.2	Gage
202	Diameter of wire	II	3.2	Gage
203	Straightness of wire	II	3.2	Gage
204	Evidence of poor workmanship	II	3.9	Visual

4.3.2.11 Pin, square loop head.				Drawing Number 9296882 Next Higher Assembly 9296868
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major	None defined			
Minor				
201	Overall length	II	3.2	Gage
202	Length to square loop	II	3.2	Gage
203	Length inside loop	II	3.2	Gage
204	Width inside loop	II	3.2	Gage
205	Diameter of wire	II	3.2	Gage
206	Protective finish missing	II	3.2	Visual
207	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.12 Spring, firing device, demolition.				Drawing Number 9296881 Next Higher Assembly 9296868
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Solid length, max	IV	3.2	Gage
102	Load at compressed length	IV	3.2	Gage
Minor				
201	End(s) not closed or not ground	II	3.2	Visual
202	Diameter of coil	II	3.2	Gage
203	Evidence of poor workmanship	II	3.9	Visual

4.3.2.13 Pin and cord assembly.				Drawing Number 9296869 Next Higher Assembly 9296868
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major	None defined			
Minor				
201	Insufficient cord length	II	3.2	Visual
202	Tie incorrect	II	3.2	Visual
203	Cotter pin missing	II	3.2	Visual
204	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.14 Firing device demo multi-purpose M142, prior to closing.				Drawing Number 9296865 Next Higher Assembly
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Any component, any assembly or instruction sheet missing	IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	II	3.9	Visual

4.3.2.15 Firing device demo multi-purpose M142.				Drawing Number 9296865 Next Higher Assembly
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Closing tape missing or insecure	IV	3.2	Visual
102	Tape ends not folded back	IV	3.2	Visual
103	Marking missing, incomplete, or illegible	IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.16 Instruction sheet.				Drawing Number 9296874 Next Higher Assembly
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Any warning statement, phrase or word not printed in proper color	IV	3.2	Visual
102	Any instruction missing, incomplete, illegible, or incorrect	IV	3.2	Visual
Minor				
201	Insufficient length of sheet	II	3.2	Visual
202	Insufficient width of sheet	II	3.2	Visual
203	Evidence of poor workmanship	II	3.9	Visual

4.3.2.17 Container, bottom.				Drawing Number 9335591 Next Higher Assembly 9296865
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Outside diameter	IV	3.2	Gage
102	Width	IV	3.2	Gage
103	Container lip not curled	IV	3.2	Visual
Minor				
201	Protective finish missing	II	3.2	Visual
202	Height to annular projection	II	3.2	Gage
203	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.18 Container, lid.				Drawing Number 9335590 Next Higher Assembly 9296865
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Large diameter	IV	3.2	Gage
102	Width	IV	3.2	Gage
Minor				
201	Length of flat head	II	3.2	Visual
202	Evidence of poor workmanship	II	3.9	Visual

4.3.2.19 Liner, for box, ammunition, M19A1.				Drawing Number 9311139 Next Higher Assembly 9311141
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major	None defined			
Minor				
201	Length	II	3.2	Scale
202	Inside width	II	3.2	Scale
203	Inside height	II	3.2	Scale
204	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.20 Box, wood, wirebound.				Drawing Number 9311140 Next Higher Assembly 9311141
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major	None defined			
Minor				
201	End boards do not project beyond edge of battens or beyond end of boards	II	3.2	Visual
202	Inside length	II	3.2	Gage
203	Inside width	II	3.2	Gage
204	Inside depth	II	3.2	Gage
205	Evidence of poor workmanship	II	3.9	Visual

4.3.2.21 Packing and marking for box, ammunition, M19A1, firing device, demolition, multi-purpose M142.				Drawing Number 9311141 Next Higher Assembly
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Any packing component missing	IV	3.2	Visual
102	Lifting strap missing on middle firing device	IV	3.2	Visual
103	Incorrect number of firing devices	IV	3.2	Visual
104	Marking missing, incomplete, or illegible	IV	3.2	Visual
Minor				
201	Box color incorrect	II	3.2	Gage
202	Box not securely closed	II	3.2	Gage
203	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.22 Packing and marking for box, wood, wirebound for boxes, ammunition M19A1 prior to closing.				Drawing Number 9311138 Next Higher Assembly
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Number of boxes incorrect	IV	3.2	Visual
102	Any packing component missing	IV	3.2	Visual
103	Any box inverted	IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	II	3.9	Visual

4.3.2.23	Packing and marking for box, wood, wirebound for boxes, ammunition M19A1 after closing.			Drawing Number 9311138 Next Higher Assembly
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Box damaged to the extent that contents are exposed or liable to become exposed	IV	3.2	Visual
102	Wire binding missing, broken or loose	IV	3.2	Visual
Minor				
201	Wire binding mislocated or improperly engaged	II	3.2	Visual
202	Contents loose	II	3.2	Visual
203	Marking misleading or unidentifiable	II	3.2	Visual
204	Metallic seal missing, unsealed or incorrectly positioned	II	3.2	Visual
205	Cleats or battens split or broken	II	3.2	Visual
206	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.2.24 Clip, retaining (if used).				Drawing Number 12589971 Next Higher Assembly 9296867
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Incorrect material	IV	3.2	Per drawing Note 2
102	Incorrect heat treatment	IV	3.2	Per drawing Note 4
103	Finish	IV	3.2	Per drawing Note 5
104	Radius (S)	IV	3.2	Gage
105	Total length	IV	3.2	Gage
106	Total length of inner clip	IV	3.2	Gage
107	Turn-up height	IV	3.2	Gage
108	Length from centerline to closing ends	IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	II	3.9	Visual

4.3.3 Dimensional control of molded parts. (The requirements of this paragraph apply to those dimensions which are considered 'as molded', and which do not undergo subsequent operations). In place of normal sampling and inspections associated with the Classification of Defects and after a curing time for the parts has been approved (see 6.3) a sample of five (5) parts (as molded) from each cavity shall be inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification, when more than one cavity is used in production. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.3.3.1 through 4.3.3.5 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the five (5) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-1916, using a VL of II for each minor defect. All cavities removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing they present objective evidence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.3.1 Nut, coupling.				Drawing Number 9296872 Next Higher Assembly 9296866
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Molding defects (see 6.5)	IV	3.2	Visual
102	Diameter of small hole	IV	3.2	Gage
103	True position of small hole with outside diameter	IV	3.2	Gage
104	True position of small hole with pitch diameter of thread	IV	3.2	Gage
Minor				
201	Outside diameter	II	3.2	Gage
202	Length of nut	II	3.2	Gage
203	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.3.2 Body, coupling.				Drawing Number 9296876 Next Higher Assembly 9296866
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Web thickness	IV	3.2	Gage
102	Molding defects (see 6.5)	IV	3.2	Visual
103	Hazard band missing	IV	3.2	Visual
104	Diameter of primer hole	IV	3.2	Gage
105	Depth of primer hole	IV	3.2	Gage
106	Diameter of flash hole	IV	3.2	Gage
107	Major diameter of thread at primer end	IV	3.2	Gage
108	Pitch diameter of thread at primer end	IV	3.2	Gage
109	Major diameter of large thread	IV	3.2	Gage
110	Pitch diameter of large thread	IV	3.2	Gage
Minor				
201	Length of knurl at primer end	II	3.2	Gage
202	Length of knurl free groove	II	3.2	Gage
203	Overall length of body	II	3.2	Gage
204	Length of thread at primer end	II	3.2	Gage
205	Length of gasket groove at primer end	II	3.2	Gage
206	Length of large thread	II	3.2	Gage
207	Length of gasket groove at large thread end	II	3.2	Gage
208	Thickness thru flash hole	II	3.2	Gage
209	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.3.3 Plate, sear.				Drawing Number 9296875 Next Higher Assembly 9296868
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Concavity	IV	3.2	Gage
102	True position of 2 holes with radius and surface B	IV	3.2	Gage
103	Thickness	IV	3.2	Gage
104	Length, surface B to end	IV	3.2	Gage
105	Angle at trigger end	IV	3.2	Gage
106	Radius for firing striker	IV	3.2	Gage
Minor				
201	Thickness of pressure tab, 2 places	II	3.2	Gage
202	Width of tab	II	3.2	Gage
203	Radius at surface B	II	3.2	Gage
204	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.3.4 Body.				Drawing Number 9296877 Next Higher Assembly 9296868
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Pitch diameter of thread	IV	3.2	Gage
102	True position of rear striker cavity with forward cavity	IV	3.2	Gage
103	True position of round loop pin hole	IV	3.2	Gage
104	True position of square loop pin hole	IV	3.2	Gage
105	True position of slot	IV	3.2	Gage
106	Width of slot	IV	3.2	Gage
107	Length of slot	IV	3.2	Gage
108	Diameter of round loop pin hole	IV	3.2	Gage
109	Diameter of square loop pin hole	IV	3.2	Gage
110	Diameter of safety pin holes	IV	3.2	Gage
111	True position of safety pin hole	IV	3.2	Gage
Minor				
201	Overall height	II	3.2	Gage
202	Diameter of screw holes	II	3.2	Gage
203	Location of wire tie-down hole	II	3.2	Gage
204	Total length	II	3.2	Gage
205	Diameter of rear striker cavity	II	3.2	Gage
206	Diameter of forward striker cavity	II	3.2	Gage
207	Evidence of poor workmanship	II	3.9	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

4.3.3.5 Container, bottom and lid.				Drawing Number 9352613 Next Higher Assembly 9296865
Category	Examination Or Test	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Outside diameter of container bottom	IV	3.2	Gage
102	Width of container bottom	IV	3.2	Gage
103	Large diameter of container lid	IV	3.2	Gage
104	Width of container lid	IV	3.2	Gage
Minor				
201	Height to annular projection of container bottom	II	3.2	Gage
202	Evidence of poor workmanship	II	3.9	Visual

4.3.4 Testing.

4.3.4.1 Functioning at normal test quantity. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the applicable requirements specified, one hundred and fifty (150) firing device assemblies with primer shall be selected from each lot for test. For test quantities, modes, requirements, and methods see Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

TABLE III. Normal test quantity.

Mode or test	Sample size	Requirement paragraph	Test method paragraph
Functioning	150 total		4.4.1
Group A			
Pull	25	3.3.1	4.4.1.1
Tension release	25	3.3.2	4.4.1.2
Pressure	25	3.3.3	4.4.1.3
Pressure release	25	3.3.4	4.4.1.4
Group B (water immersion)			
Pull	10	3.5, 3.3.1	4.4.2, 4.4.1.1
Pressure	10	3.5, 3.3.3	4.4.2, 4.4.1.3
Group C (cold temp.)			
Pull	10	3.6, 3.3.1	4.4.3, 4.4.1.1
Pressure	10	3.6, 3.3.3	4.4.3, 4.4.1.3
Pressure release	10	3.6, 3.3.4	4.4.3, 4.4.1.4
Primer check test (if applicable)	900	3.7	4.4.4
Note: Accept/reject criteria shall be in accordance with 4.4.4.			

The lot shall be rejected if during test, a critical defect occurs; or if four (4) or more major functional defects occur; or if more than two (2) major functional defects occur in any one (1) firing mode (see 4.4.1); or if any device fails to meet the requirements (see 4.4.1).

4.3.4.2 Functioning at reduced test quantity. After three (3) consecutive lots have complied with the acceptance criteria of 4.3.4.1, sixty (60) firing device assemblies with primer shall be selected from each lot for test. For test quantities, modes, requirements and methods see Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

TABLE IV. Reduced test quantity.

Mode or test	Sample size	Requirement paragraph	Test method paragraph
Functioning	60 total		4.4.1
Group A			
Pull	12	3.3.1	4.4.1.1
Tension release	12	3.3.2	4.4.1.2
Pressure	12	3.3.3	4.4.1.3
Pressure release	12	3.3.4	4.4.1.4
Group B (water immersion)			
Pull	6	3.5, 3.3.1	4.4.2, 4.4.1.1
Group C (cold temp.)			
Pressure	6	3.6, 3.3.3	4.4.3, 4.4.1.3
Primer check test (if applicable)	900	3.7	4.4.4
Note: Accept/reject criteria shall be in accordance with 4.4.4.			

The lot shall be rejected if during test, a critical defect occurs; or if three (3) or more major funcional defects occur; or if any device fails to meet the requirements (see 4.4.1). After rejection, sampling shall return to 4.3.4.1.

4.4 Test methods and procedures.

4.4.1 Functioning. During function testing the following defects are defined as Critical and Major:

TABLE V. Functional defect definitions.

Critical functional defect 1	Firing pin prematurely releases during setup, prior to removal of the safety pin
Major functional defect 101	Device fails to function through the blasting cap
Major functional defect 102	Device fails to function in accordance with Pull force requirements
Major functional defect 103	Device fails to function in accordance with Tension release force requirements
Major functional defect 104	Device fails to function in accordance with Pressure force requirements
Major functional defect 105	Device fails to function in accordance with Pressure release force requirements

4.4.1.1 Pull. Secure switch to a fixed object with nails, screws or wire. Screw coupling base with attached M7 Non-Electric Blasting Cap or alternative test cap 9349653 into the switch. Attach trip wire to hole in sear plate, remove pin from a safe distance by means of a length of wire or cord. Apply the specified pull force in a direction perpendicular to sear plate, until

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

functioning occurs as indicated by detonation of blasting cap. The force required to function the item shall be observed and recorded.

4.4.1.2 Tension release. Secure switch to a fixed object with nails, screws or wire. Screw coupling base with attached M7 Non-Electric Blasting Cap or alternative test cap 9349653 into the switch. Fit tension release attachment and loop end of trip wire over the curved neck. Apply the specified tension to the trip wire and observe for compliance. Remove the pin with the round loop. Then carefully remove the safety pin (cotter pin) from a safe distance by means of a length of wire or cord. The trip wire shall then be released by cutting or flipping off a pin while maintaining the specified tension.

4.4.1.3 Pressure. Secure switch to a fixed object with nails, screws or wire. Screw coupling base with attached M7 Non-Electric Blasting Cap or alternative test cap 9349653 into the switch. Apply a load, not heavy enough to activate the switch, on the sear plate platform. Remove pin with square loop and carefully remove safety pin from a safe distance by means of a length of wire or cord. Apply the specified pressure to the sear plate platform until functioning occurs as indicated by detonation of blasting cap. The force required to function the item shall be observed and recorded.

4.4.1.4 Pressure release. Secure switch to a fixed object with nails, screws or wire. Screw coupling base with attached M7 Non-Electric Blasting Cap or alternative test cap 9349653 into the switch. A preload of four (4) pounds shall be applied to the sear plate platform. Remove the pin with the round loop. Then carefully remove the safety pin (cotter pin) from a safe distance by means of a length of wire or cord. Gradually reduce the load on the sear plate platform while observing the gage to determine for the force at which the item functions. Record force at which item functions.

4.4.2 Water immersion. Secure switch to a fixed object with nails, screws or wire. Screw coupling base with attached M7 Non-Electric Blasting Cap or alternative test cap 9349653 into the switch. Immerse in water to a covering depth of 6.0 inches. Proceed with pull test, method 4.4.1.1, and pressure (push), method 4.4.1.3.

4.4.3 Cold temperature. Condition firing devices, with coupling base with attached M7 Non-Electric Blasting Cap or alternative test cap 9349653 screwed in, at minus 50°F for not less than sixteen (16) hours. Remove one at a time and proceed with pull test, method 4.4.1.1, pressure (push), method 4.4.1.3 and pressure release, method 4.4.1.4 as quickly as possible.

4.4.4 Check test for possible deterioration of primers. Primers that fail to meet primer deterioration requirements must satisfactorily pass the check test for deterioration specified in MIL-P-20444 immediately before lot is assembled into the coupling body. The check test shall be performed by the contractor (see 6.4) prior to assembling the primers into the coupling bodies. The test shall be performed as necessary to meet the applicable sensitivity requirements in accordance with MIL-P-20444.

4.4.5 Body, coupling web. The specified pressure shall be applied from the output end toward the primer cavity for the required time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Materials, components, and assemblies. The materials, components, and assemblies shall comply with all the requirements of Drawings 9311138, 9311141, 9311140, 9311139, and all subsidiary drawings.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The firing device, multi-purpose, M142, a single firing device capable of being set for pull, tension (pull) release, pressure (push) or pressure release is intended for use in activating demolition blocks and other demolition devices and in booby-trapping.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Requirements for Ammunition Lot Numbering, if applicable.
d. Requirement for acceptance inspection equipment (AIE) designs.
e. Certificate of Conformance for each lot or shipment of product.

6.3 Curing time. In establishing the curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. minutes) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time should be submitted to the Government for approval. If any new change in material, cycle time, a reworked cavity, or a new cavity is used, a new curing time should be established and re-submitted for approval. Data used to establish curing time should be sent to U.S. Army CCDC - AC, Picatinny Arsenal, New Jersey 07806-5000, ATTN: FCDD-ACE-QMG.

6.4 Molding defects. Molding defects are defined as incomplete form, blisters, excessive flash, foreign materials, or any other defect that would make the part unserviceable.

6.5 Automated Acceptance Inspection Equipment (AAIE). AAIE is defined as Automated Acceptance Inspection Equipment. See MIL-A-70625 for additional guidance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.

6.6 Critical classification rationale.

TABLE VI. Critical classification rationale.

Paragraph	Critical callouts	Critical rationale
4.3.2.7	C1 – Safety pin not in place or legs not bent	Missing or incorrectly assembled safety devices results in catastrophic risk severity and warrants a critical classification.
4.3.2.7	C2 – Firing pin released against safety pin with all components (pins) present	Prematurely released firing pins may cause unintentional functioning during setup which results in catastrophic risk severity and warrants a critical classification.
Table V	C1 – Firing pin prematurely releases during setup, prior to removal of the safety pin	Prematurely released firing pins may cause unintentional functioning during setup which results in catastrophic risk severity and warrants a critical classification.

6.7 Subject term (key word) listing.
Initiator
Booby trap defeat

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Army-AR

Preparing Activity:
Army-AR
(Project 1375-2020-009)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:10Z
Check the source to verify that this is the current version before use.